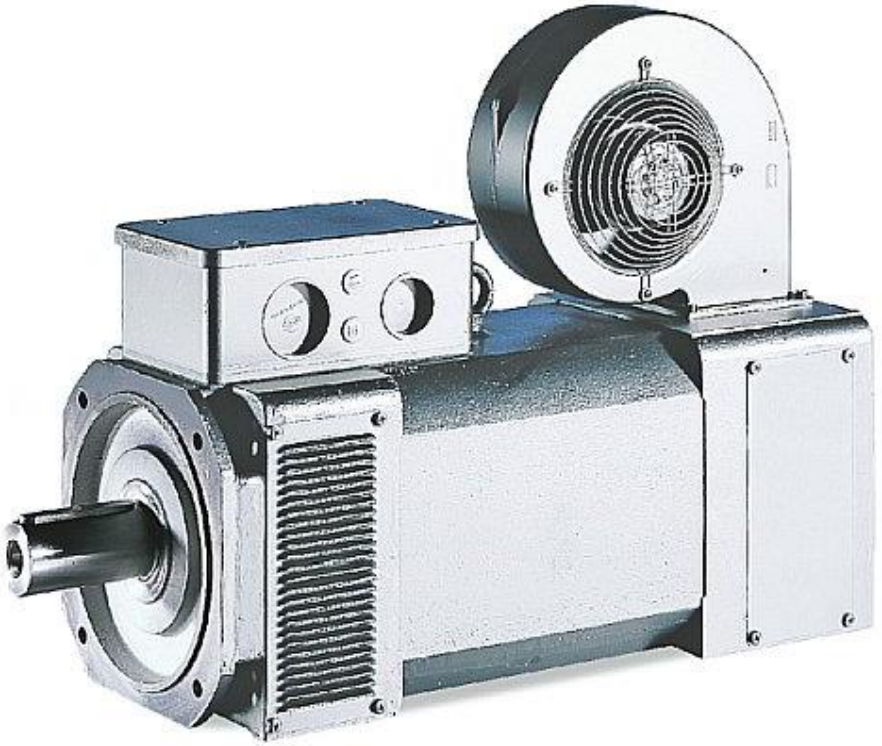
EEE 2421
Electrical Drives and Instrumentation





Developed by: Mohammad Mahmudul Hasan Tareq

Electrical Drives and Instrumentation

Principle and Application of PV

What is PV Cell or Solar Cell?

- A solar cell (also known as a photovoltaic cell or PV cell) is defined as an electrical device that converts light energy into electrical energy through the photovoltaic effect

- A solar cell is basically a p-n junction diode

- Solar cells are a form of photoelectric cell, defined as a device whose electrical characteristics – such as current, voltage or resistance – vary when exposed to light

- The common single junction silicon solar cell can produce a maximum open-circuit voltage of approximately 0.5 to 0.6 volts

What is PV Cell or Solar Cell?

- Individual solar cells can be combined to form modules commonly known as solar panels



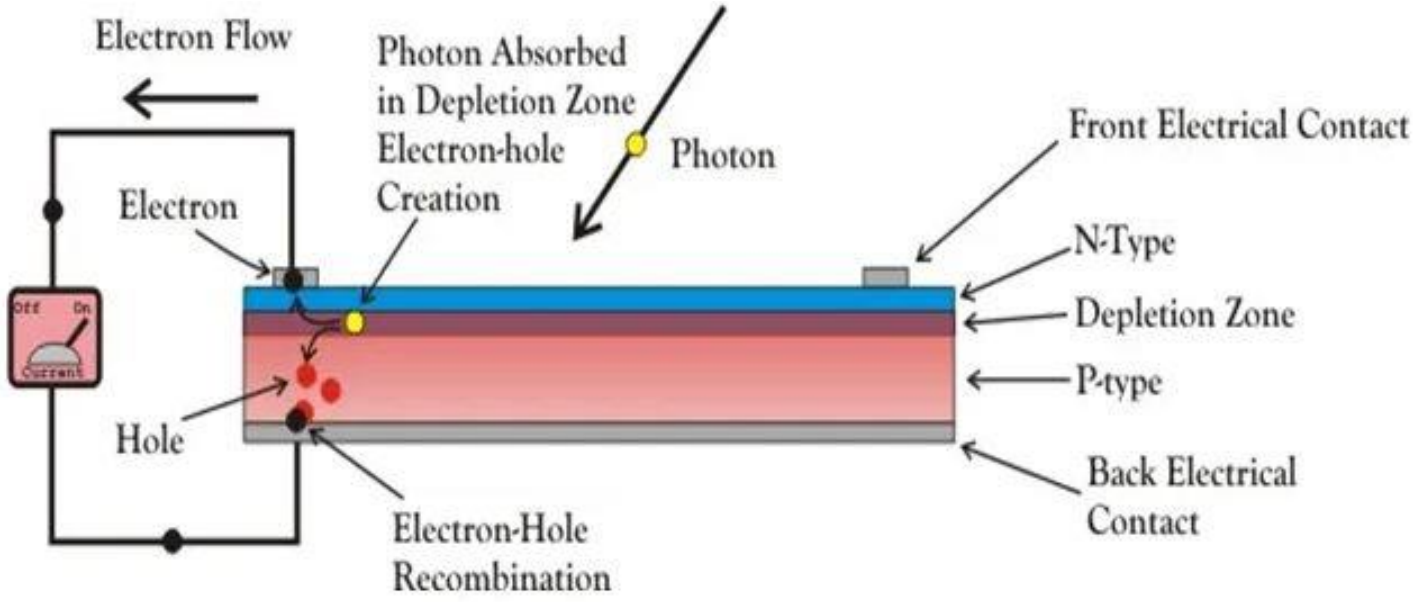
Construction of Solar Cell

- A solar cell is basically a junction diode, although its construction it is little bit different from conventional p-n junction diodes
- A very thin layer of n-type semiconductor is grown on a relatively thicker p-type semiconductor
- We then apply a few finer electrodes on the top of the p-type semiconductor layer
- These electrodes do not obstruct light to reach the thin p-type layer
- Just below the n-type layer there is a p-n junction

Construction of Solar Cell

- We encapsulate the entire assembly by thin glass to protect the solar cell from any mechanical shock

- We also provide a current collecting electrode at the bottom of the p-type layer



Working Principle of Solar Cell

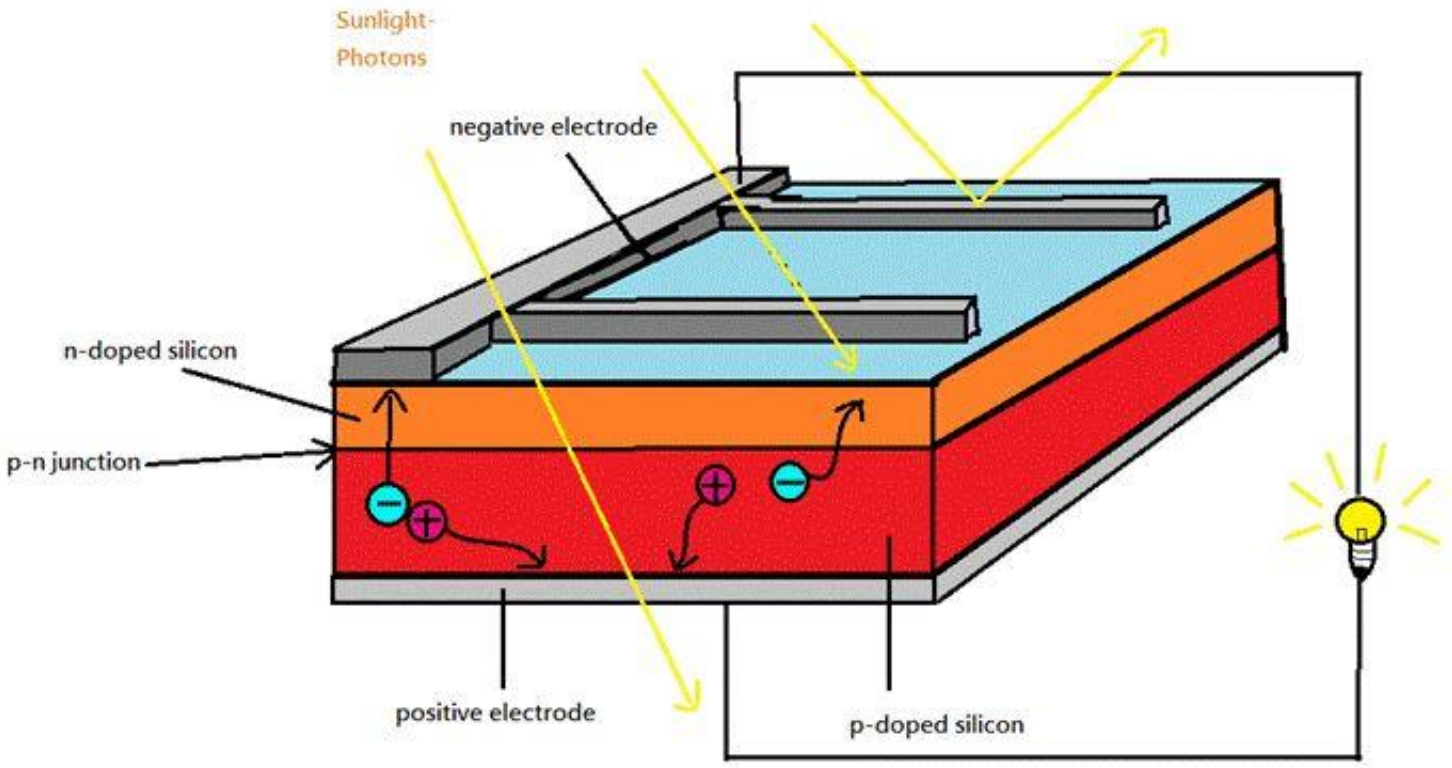
- When light reaches the p-n junction, the light photons can easily enter in the junction, through very thin n-type layer
- The light energy, in the form of photons, supplies sufficient energy to the junction to create a number of electron-hole pairs
- The incident light breaks the thermal equilibrium condition of the junction
- The free electrons in the depletion region can quickly come to the n-type side of the junction
- Similarly, the holes in the depletion can quickly come to the p-type side of the junction

Working Principle of Solar Cell

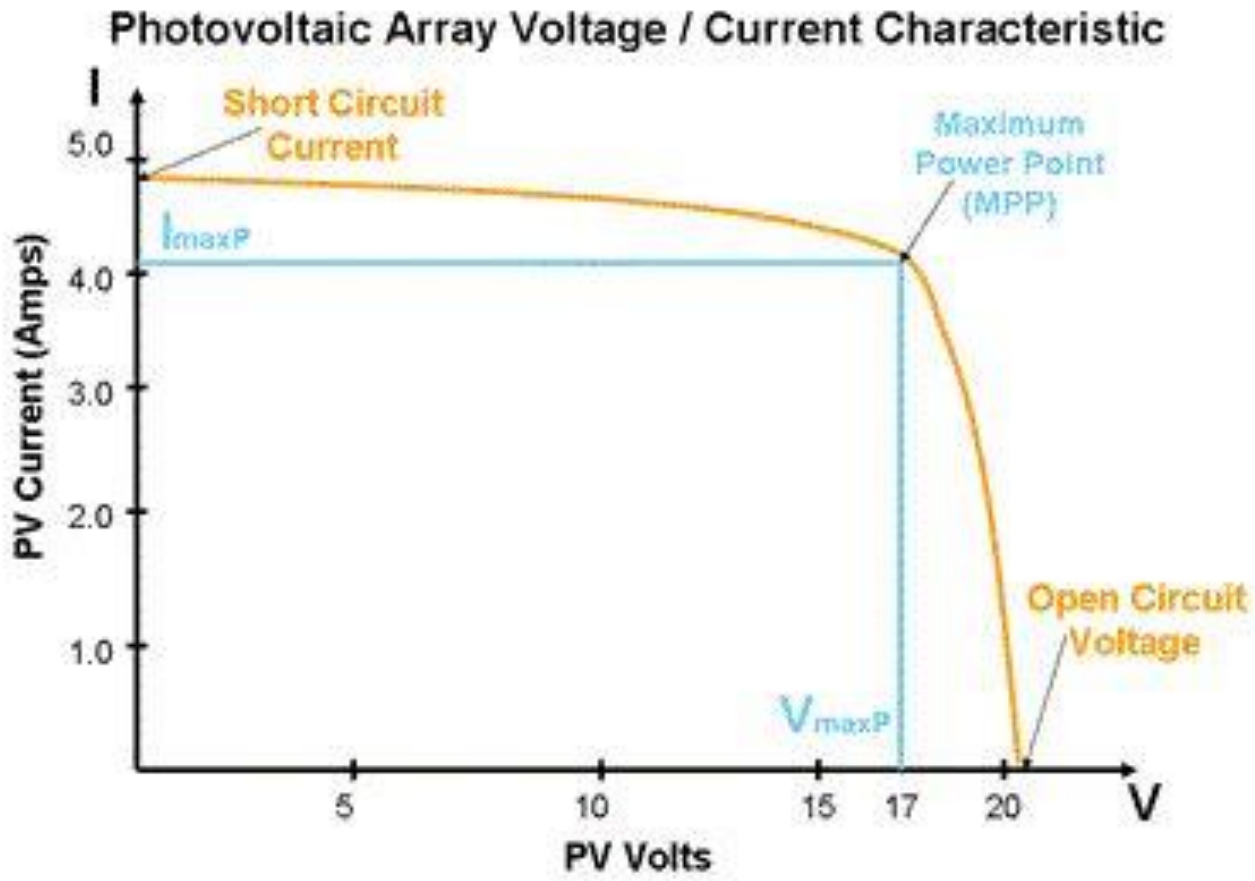
- Once, the newly created free electrons come to the n-type side, cannot further cross the junction because of barrier potential of the junction
- Similarly, the newly created holes once come to the p-type side cannot further cross the junction became of same barrier potential of the junction
- As the concentration of electrons becomes higher in one side, i.e. n-type side of the junction and concentration of holes becomes more in another side, i.e. the p-type side of the junction, the p-n junction will behave like a small battery cell
- A voltage is set up which is known as photo voltage

Working Principle of Solar Cell

- If we connect a small load across the junction, there will be a tiny current flowing through it



V-I Characteristics of a Photovoltaic Cell



Materials Used in Solar Cell

- The materials which are used for this purpose must have band gap close to 1.5 ev
- Commonly used materials are-

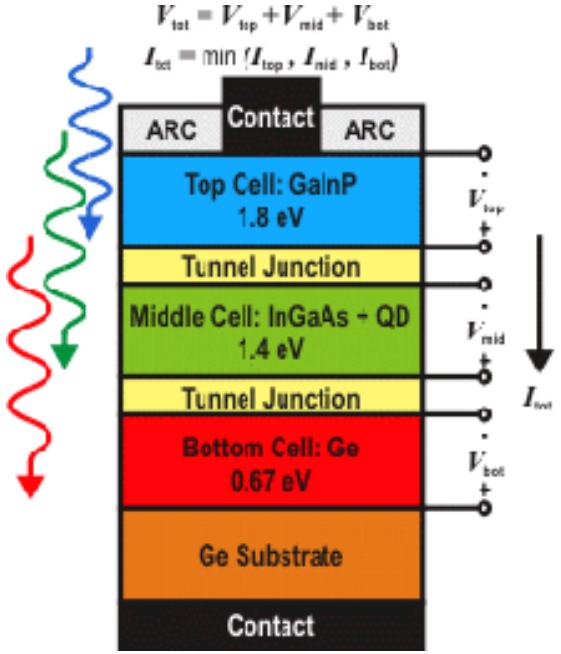
1. Silicon
2. GaAs
3. CdTe
4. $CuInSe_2$

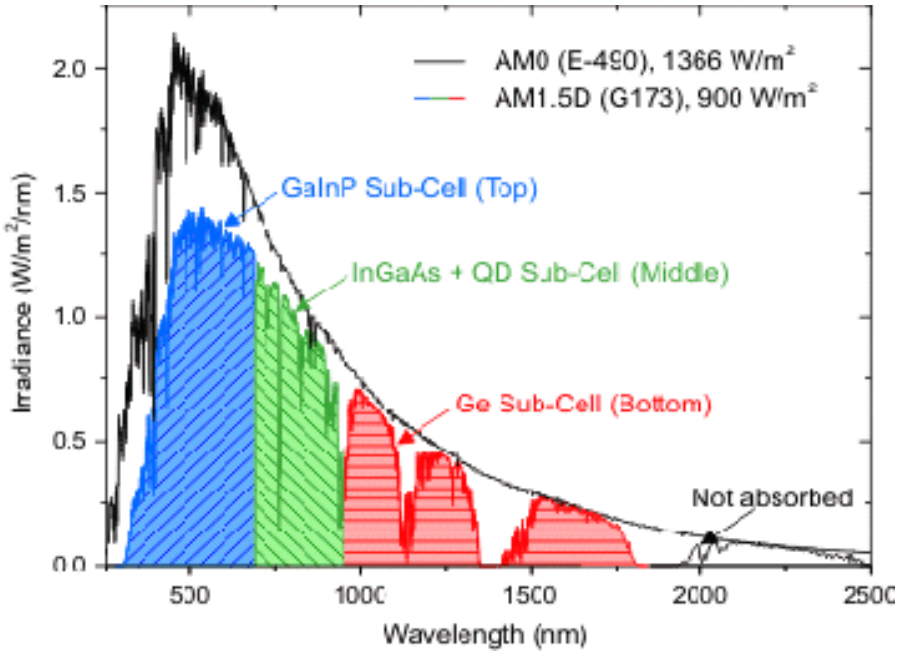
Criteria for Materials to be Used in Solar Cell

1. Must have band gap from 1 ev to 1.8 ev
2. It must have high optical absorption
3. It must have high electrical conductivity
4. The raw material must be available in abundance and the cost of the material must be low

Criteria for Materials to be Used in Solar Cell

$$E(eV) = \frac{1.24}{\lambda(\mu m)}$$





Advantages of Solar Cell

Advantages of Solar Cell:

1. No pollution associated with it
2. It must last for a long time
3. No maintenance cost

Dis - advantages of Solar Cell

Dis - advantages of Solar Cell:

1. It has high cost of installation
2. It has low efficiency
3. During cloudy day, the energy cannot be produced and also at night we will not get solar energy

Applications of Solar Cell

Solar Farms:

- Many acres of PV panels can provide utility-scale power—from tens of megawatts to more than a GW of electricity. These large systems, using fixed or sun-tracking panels, feed power into municipal or regional grids

Remote Locations:

- It is not always cost-effective, convenient or even possible to extend power lines to locations where electricity is needed. PV can be the solution—for rural homes, villages in developing nations, lighthouses, offshore oil platforms, desalination plants, and remote health clinics

Applications of Solar Cell

Stand – Alone Power:

- In urban or remote areas, PV can power stand-alone devices, tools and meters. PV can meet the need for electricity for parking meters, temporary traffic signs, emergency phones, radio transmitters, water irrigation pumps, stream-flow gauges, remote guard posts, lighting for roadways and more

Power in Space:

- From the beginning, PV has been a primary power source for Earth-orbiting satellites. High-efficiency PV has supplied power for ventures such as the International Space Station and surface rovers on the Moon and Mars and it will continue to be an integral part of space and planetary exploration

Applications of Solar Cell

Building Related Needs:

- In buildings, PV panels mounted on roofs or ground can supply electricity. PV material can also be integrated into a building's structure as windows, roof tiles to serve a dual purpose. In addition, awnings and parking structures can be covered with PV to provide shading and power

Military Uses:

- Lightweight, flexible thin-film PV can serve applications in which portability or ruggedness are critical. Soldiers can carry lightweight PV for charging electronic equipment in the field or at remote bases

Applications of Solar Cell

Transportation:

- PV can provide auxiliary power for vehicles such as cars and boats. Automobile sunroofs can include PV for onboard power needs or trickle-charging batteries. Lightweight PV can also conform to the shape of airplane wings to help power high-altitude aircraft

Electrical Drives and Instrumentation

Principle and Application of Piezoelectric

What is Piezoelectric Transducer?

- A Piezoelectric transducer is an electrical transducer which can convert any form of physical quantity into an electrical signal, which can be used for measurement
- An electrical transducer which uses properties of piezoelectric materials for conversion of physical quantities into electrical signals is known as a piezoelectric transducer
- Piezoelectric materials exhibit the property of piezoelectricity, according to which on the application of any type of mechanical stress or strain leads to the generation of an electric voltage proportional to the applied stress
- This produced electric voltage can be measured using voltage measuring instruments to calculate the value of stress or strain applied to the material

Types of Piezoelectric Materials

- Naturally Available Ones: Quartz, Rochelle salt, Topaz, Tourmaline-group minerals and some organic substances as silk, wood, enamel, bone, hair, rubber, dentin

- Artificially manufactures piezoelectric materials: Polyvinylidene difluoride, PVDF or PVF_2, Barium titanate, Lead titanate, Lead zirconate titanate (PZT), Potassium niobate, Lithium niobate, Lithium tantalate, and other lead-free piezoelectric ceramics

- Not all piezoelectric materials can be used in piezoelectric transducers

- There are certain requirements to be met by the piezoelectric materials to be used as transducers

Types of Piezoelectric Materials

- The materials used for measurement purpose should have frequency stability, high output values, insensitive to the extreme temperature and humidity conditions and which can be available in various shapes or should be flexible to be manufactured into various shapes without disturbing their properties

- Quartz is a highly stable crystal which is naturally available but it has small output levels

- Slowly varying parameters can be measured with quartz

- Rochelle salt gives the highest output values but it is sensitive to environmental conditions and cannot be operated above 1150F

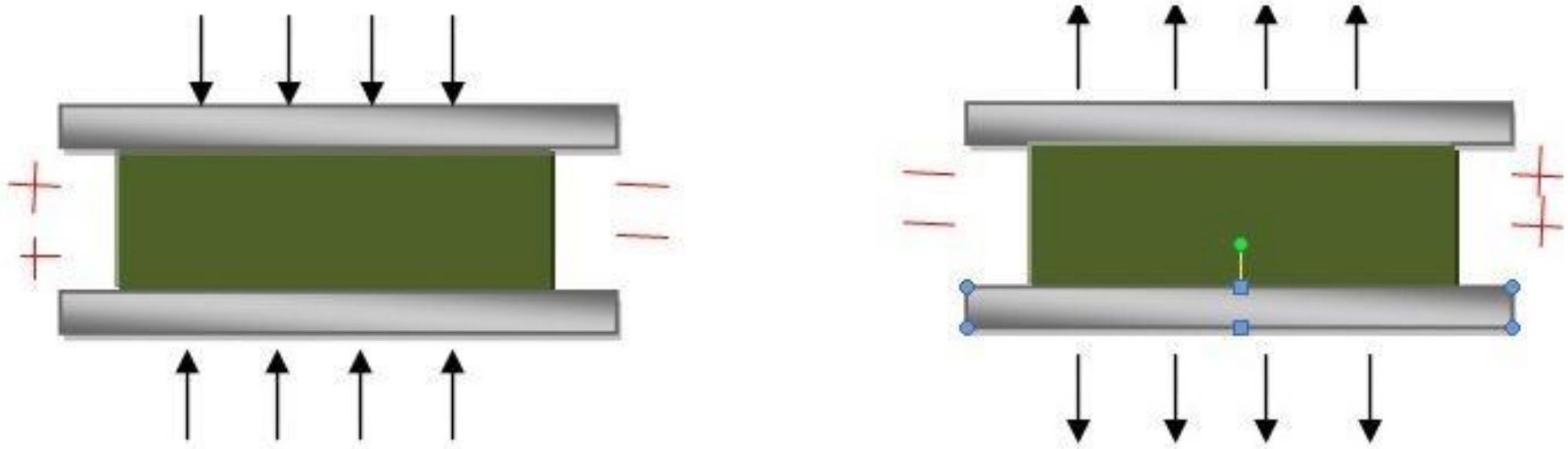
Piezoelectric Transducer Working Principle

- Piezoelectric Transducer works with the principle of piezoelectricity
- The faces of piezoelectric material, usual quartz, is coated with a thin layer of conducting material such as silver
- When stress has applied the ions in the material move towards one of the conducting surface while moving away from the other
- This results in the generation of charge
- This charge is used for calibration of stress

Piezoelectric Transducer Working Principle

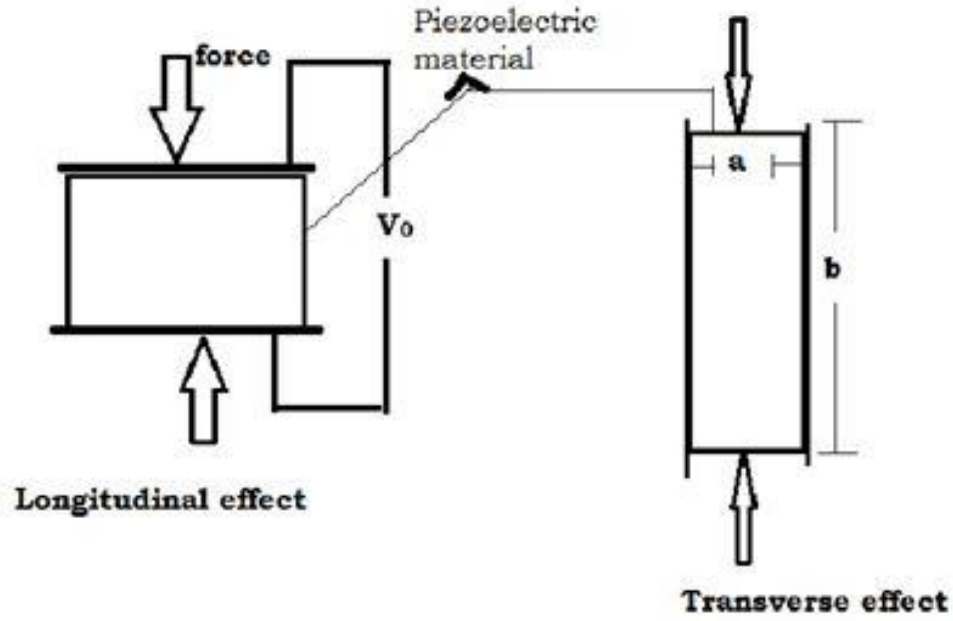
- The polarity of the produced charge depends upon the direction of the applied stress

- Stress can be applied in two forms as Compressive stress and Tensile stress as shown below



Piezoelectric Transducer Formula

- The orientation of the crystal also effects the amount of voltage generated
- Crystal in a transducer can be arranged in longitudinal position or transverse position



Piezoelectric Transducer Formula

Longitudinal and Transverse Effect:

- In the longitudinal effect, the charge generated is given by

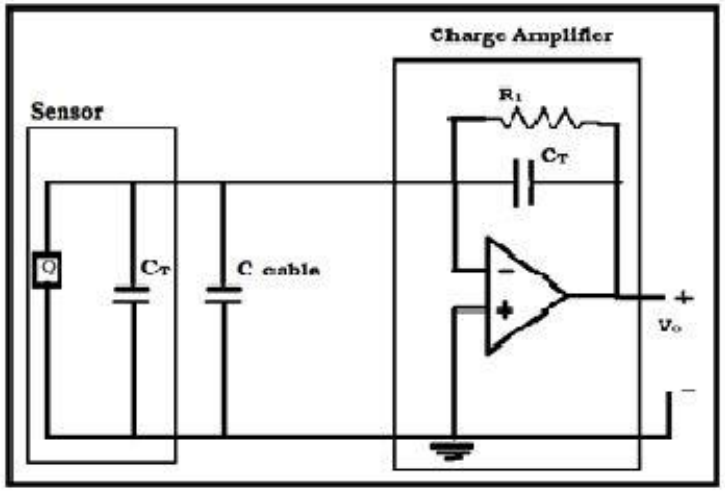
$$Q = F \times d$$

- Where F is the applied force, d is the piezoelectric coefficient of the crystal
- Piezoelectric coefficient d of quartz crystal is around 2.3×10^{-12} C/N
- In the transverse effect, the charge generated is given by

$$Q = F \times d \times (b/a)$$

Piezoelectric Transducer Circuit

- The working of a basic piezoelectric transducer can be explained by the below figure



- Here quartz crystal coated with silver is used as a sensor to generate a voltage when stress is applied on it

- A charge amplifier is used to measure the produced charge without dissipation

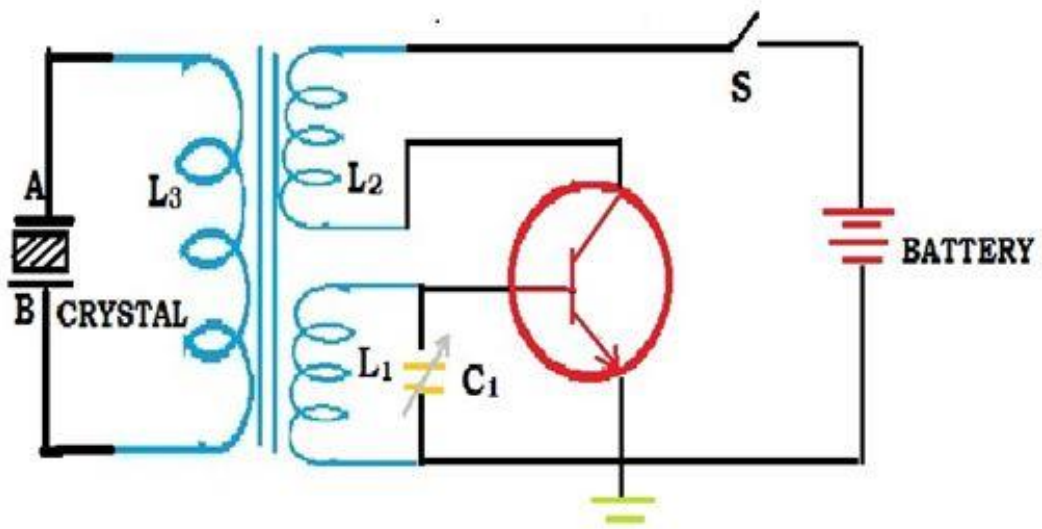
Piezoelectric Transducer Circuit

- To draw very low current the resistance R1 is very high
- The capacitance of the lead wire that connects the transducer and piezoelectric sensor also affects the calibration
- So the charge amplifier is usually placed very near to the sensor
- So in a piezoelectric transducer when mechanical stress is applied a proportional electric voltage is generated which is amplified using charge amplifier and used for calibration of applied stress

Piezoelectric Ultrasonic Transducer

- The ultrasonic piezoelectric transducer works on the principle of the reverse piezoelectric effect

- In this effect when electricity is applied to a piezoelectric material, it undergoes physical deformations proportional to applied charge



Piezoelectric Ultrasonic Transducer

- Here, the quartz crystal is placed between two metal plates A and B which are connected to the primary L3 of the transformer
- The primary of the transformer is inductively coupled to the electronic oscillator
- The coils L1 and L2, which forms secondary of the transformer, are connected to the electronic oscillator
- When the battery is turned ON the oscillator produces high-frequency alternating voltage pulses with a frequency $f = \frac{1}{2\pi\sqrt{L1C1}}$
- Due to this, an e.m.f is induced in L3 which is transferred to the quartz crystal through plates A and B

Piezoelectric Ultrasonic Transducer

- Due to reverse piezoelectric effect the crystal starts contracting and expanding alternatively thus creating mechanical vibrations
- Resonance takes place when the frequency of the electronic oscillator is equal to the natural frequency of the quartz
- At this point, quartz produces longitudinal ultrasonic waves of large amplitude

Piezoelectric Transducer Application

- As piezoelectric materials cannot measure static values these are primarily used for measuring surface roughness, in accelerometers and as a vibration pickup
- They are used in seismographs to measure vibrations in rockets
- In strain gauges to measure force, stress, vibrations etc.
- Used by automotive industries to measure detonations in engines
- These are used in ultrasonic imaging in medical applications

Advantages of Piezoelectric Transducer

Advantages:

- These are active transducer i.e. they don't require external power for working and are therefore self-generating
- The high-frequency response of these transducers makes a good choice for various applications

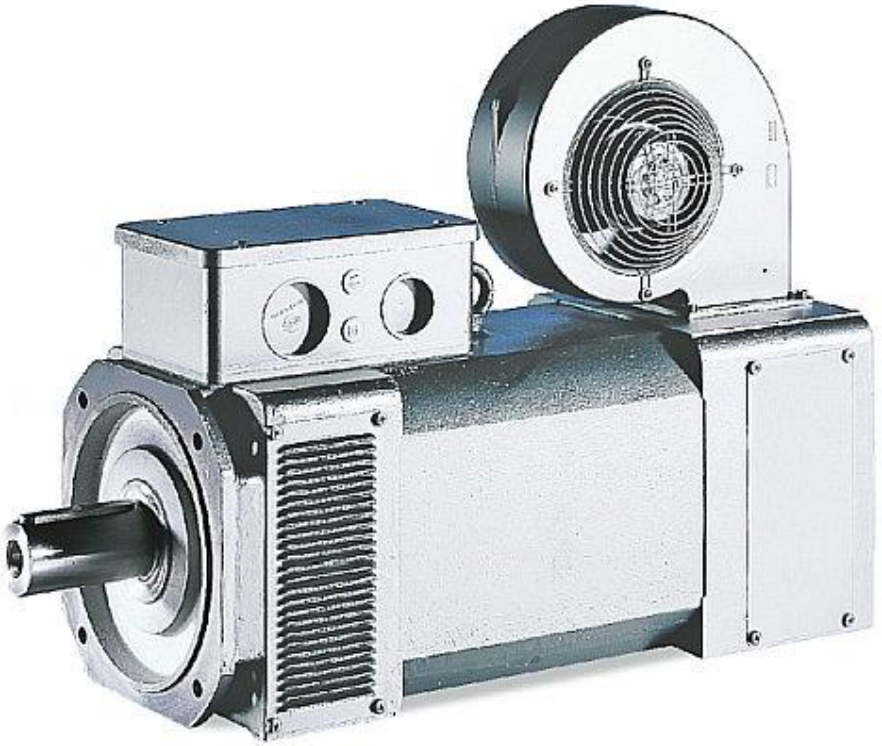
Limitations of Piezoelectric Transducer

Limitations:

- Temperature and environmental conditions can affect the behavior of the transducer
- They can only measure changing pressure hence they are useless while measuring static parameters


Thank
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EEE 2421
Electrical Drives and Instrumentation





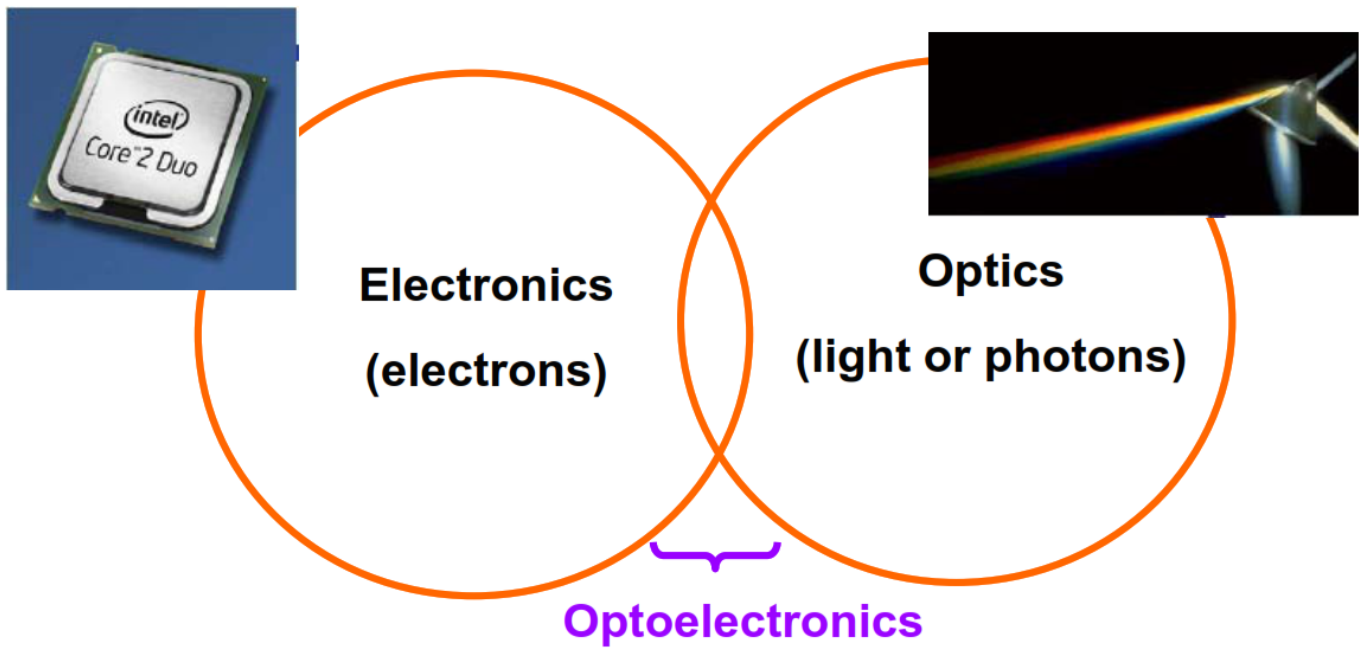
Course Teacher: Mohammad Mahmudul Hasan Tareq

Electrical Drives and Instrumentation

Principle and Application of Optoelectronic Transducer

What Did the Word “Opto-Electronics” Mean?

- **Optoelectronics** is the study and application of electronic devices that interact with light



What is Optoelectronics?

- Optoelectronics deals with the study and application of electronic devices that interacts with light which might be detection of light, its creation and exploitation for several purposes
- This includes Gamma rays, X ray, Ultraviolet, Infrared and visible light
- It also encompasses the study, design and manufacture of hardware apparatus that facilitate the conversion of electricity into photon signals
- Optoelectronic devices are primarily transducers i.e. they can convert one energy form to another

What is Optoelectronics?

- These devices produce light by expending electrical energy
- They can also detect light and transform light signals to electrical signals for processing by a computer
- Optoelectronics makes use of the quantum mechanical effect of light
- This property is used mainly in the materials that are used in the manufacture of semiconductors

Types of Optoelectronics Device

Optoelectronics are classified into different types such as:

1. Photodiode
2. Solar Cells
3. Light Emitting Diodes
4. Optical Fiber
5. Laser Diodes

Types of Optoelectronics Device

Photo Diode:

- A photo diode is a semiconductor light sensor that generates a voltage or current when light falls on the junction
- It consists of an active P-N junction, which is operated in reverse bias
- When a photon with plenty of energy strikes the semiconductor, an electron or hole pair is created
- The electrons diffuse to the junction to form an electric field
- This electric field across the depletion zone is equal to a negative voltage across the unbiased diode

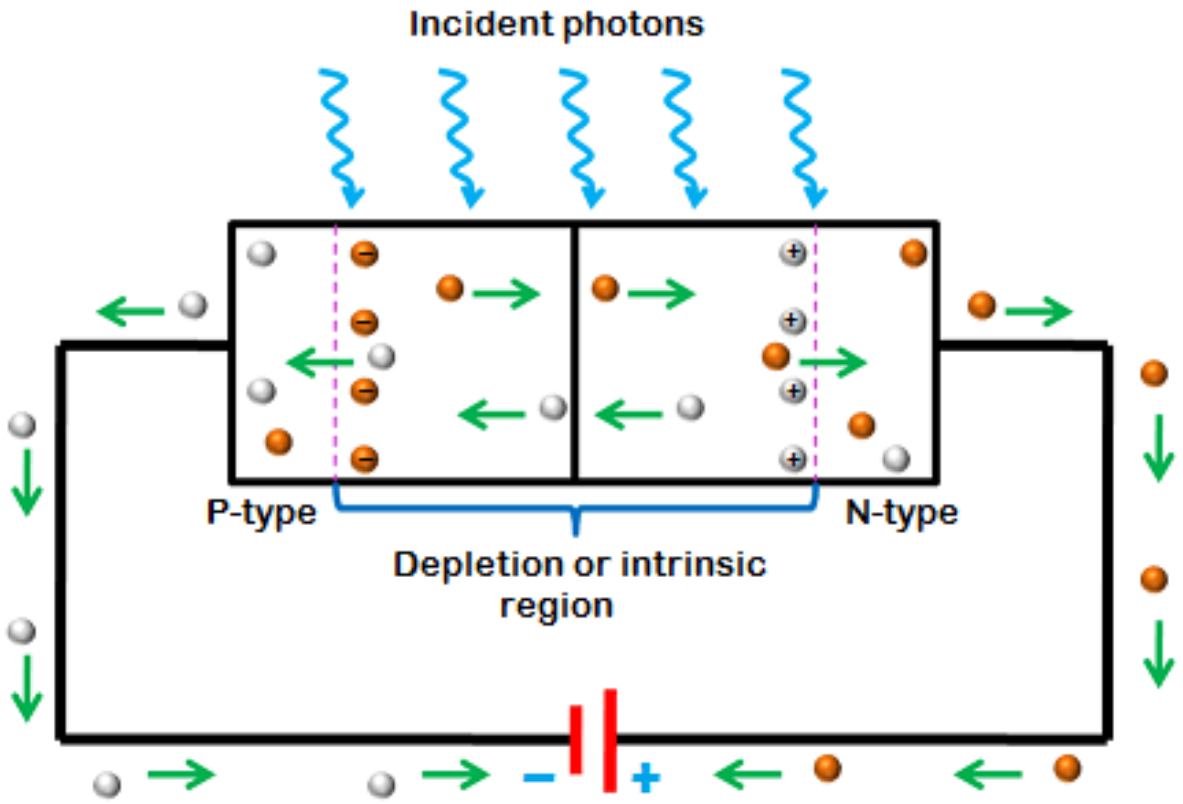
Types of Optoelectronics Device

Photo Diode:

- This method is also known as the inner photoelectric effect
- This device can be used in three modes: photovoltaic as a solar cell, forward biased as an LED and reverse biased as a photo detector
- Photodiodes are used in many types of circuits and different applications such as cameras, medical instruments, safety equipment, industries, communication devices and industrial equipment

Types of Optoelectronics Device

Photo Diode:



Types of Optoelectronics Device

Solar Cells:

- A solar cell or photo-voltaic cell is an electronic device that directly converts sun's energy into electricity
- When sunlight falls on a solar cell, it produces both a current and a voltage to produce electric power
- Sunlight, which is composed of photons, radiates from the sun
- When photons hit the silicon atoms of the solar cell, they transfer their energy to lose electrons and then, these high-energy electron flow to an external circuit

Types of Optoelectronics Device

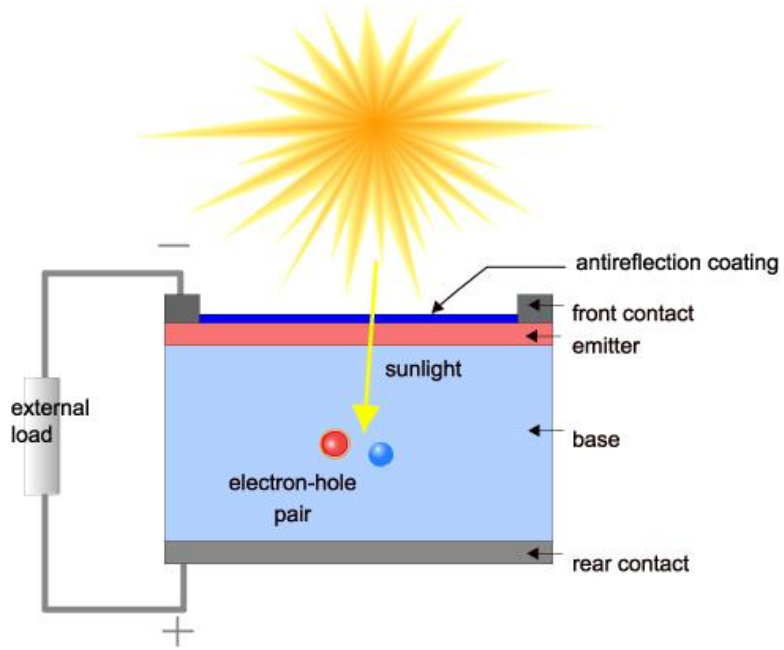
Solar Cells:

- The solar cell is composed of two layers which are struck together
- The first layer is loaded with electrons, so these electrons are ready to jump from the first layer to the second layer
- The second layer has some electrons taken away and therefore, it is ready to take more electrons
- The advantages of solar cells are that, there is no fuel supply and cost problem
- These are very dependable and require little maintenance

Types of Optoelectronics Device

Solar Cells:

- The solar cells are applicable in rural electrification, telecommunication systems, ocean navigation aids, electric power generation system in space and remote monitoring and control systems



Types of Optoelectronics Device

Light-Emitting Diodes:

- Light – emitting diode is a P-N semiconductor diode in which the recombination of electrons and holes yields a photon
- When the diode is electrically biased in the forward direction, it emits incoherent narrow spectrum light
- When a voltage is applied to the leads of the LED, the electrons recombine with the holes within the device and release energy in the form of photons
- This effect is called as electroluminescence. It is the conversion of electrical energy into light

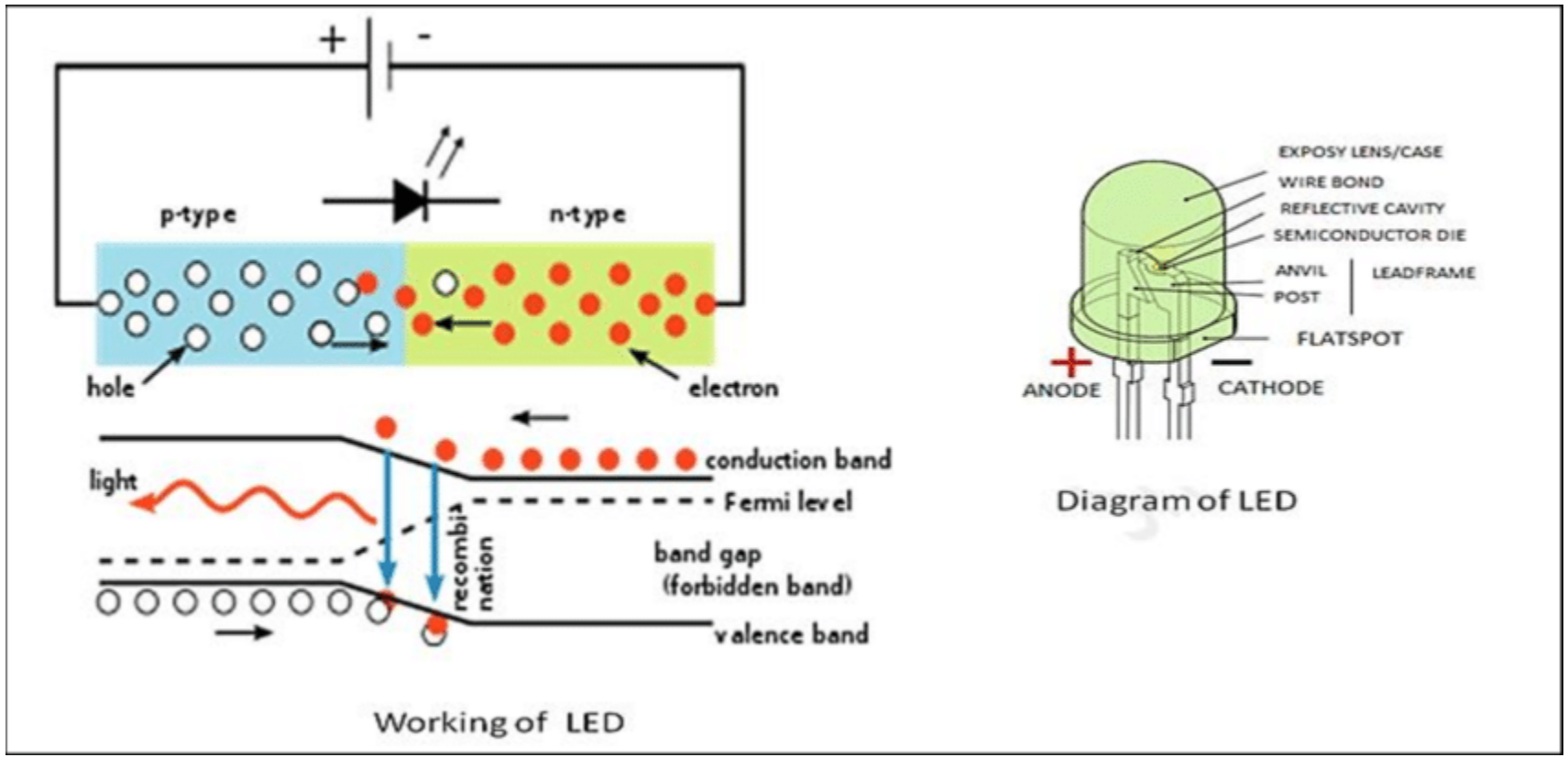
Types of Optoelectronics Device

Light-Emitting Diodes:

- The color of the light is decided by the energy band gap of the material
- The usage of LED is advantageous as it consumes less power and produces less heat
- LEDs last longer than incandescent lamps
- LEDs could become the next generation of lighting and used anywhere like in indication lights, computer components, medical devices, watches, instrument panels, switches, fiber-optic communication, consumer electronics, household appliances, etc.

Types of Optoelectronics Device

Light-Emitting Diodes:



Types of Optoelectronics Device

Optical Fiber:

- An optical fiber or optic fiber is a plastic and transparent fiber made of plastic or glass
- It is somewhat thicker than a human hair
- It can function as a light pipe or waveguide to transmit light between the two ends of the fiber
- Optical fibers usually include three concentric layers: a core, a cladding and a jacket

Types of Optoelectronics Device

Optical Fiber:

- The core, a light transmitting region of the fiber, is the central section of the fiber, which is made of silica
- Cladding, the protective layer around the core, is made of silica
- This creates an optical waveguide that limits the light in the core by total reflection at the interface of the core-cladding
- Jacket, the non-optical layer around the cladding, typically consists of one or more layers of a polymer that protect the silica from the physical or environmental damage

Types of Optoelectronics Device

Optical Fiber:

- Along with the fiber-optic cable, jackets are available in different colors
- These colors allow the recognition of the fiber-optic cable and the type of cable one is dealing with
- For example, an orange-color cable clearly indicates a single-mode fiber, while a yellow one indicates a multimode fiber
- In the single-mode fiber, one mode propagates and the light rays travel straight through the cable

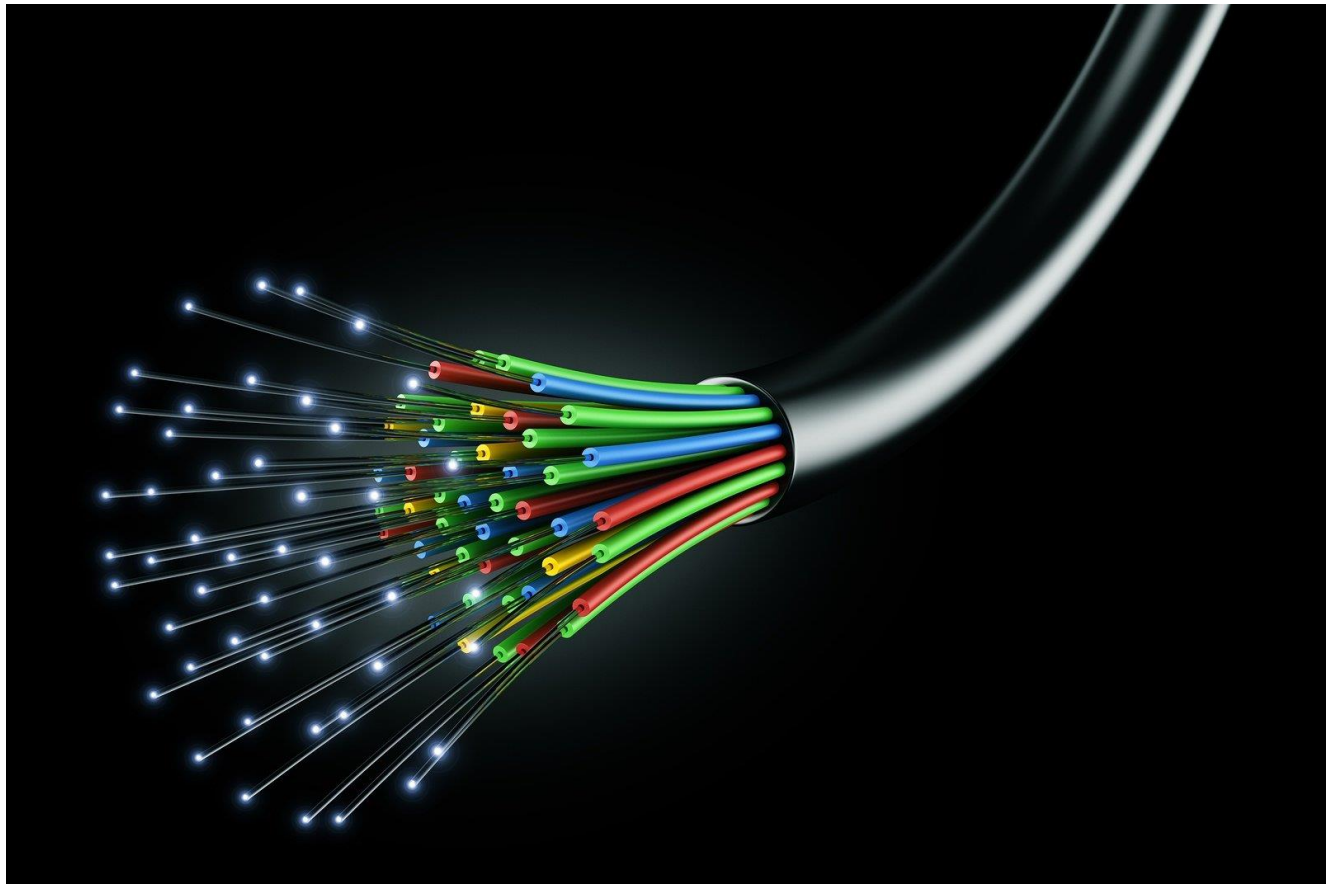
Types of Optoelectronics Device

Optical Fiber:

- In a multimode cable, the light rays travel through the cable following different modes
- These cables are used in telecommunications, sensors, fiber lasers, bio-medicals and in many other industries
- The advantages of using optical-fiber cable include their higher bandwidth, less signal degradation, weightlessness and thinness than a copper wire, cost-effectiveness, flexibility and hence they are used in medical and mechanical imaging systems

Types of Optoelectronics Device

Optical Fiber:



Types of Optoelectronics Device

Laser Diodes:

- Laser (light amplification by stimulated emission of radiation) is a source of highly monochromatic, coherent and directional light
- It operates under stimulated emission condition
- The function of a laser diode is to convert electrical energy into light energy like infrared diodes or LEDs
- The most common lasers used are injection lasers or semiconductor lasers
- The semiconductor laser changes from other lasers like solid, liquid and gas lasers

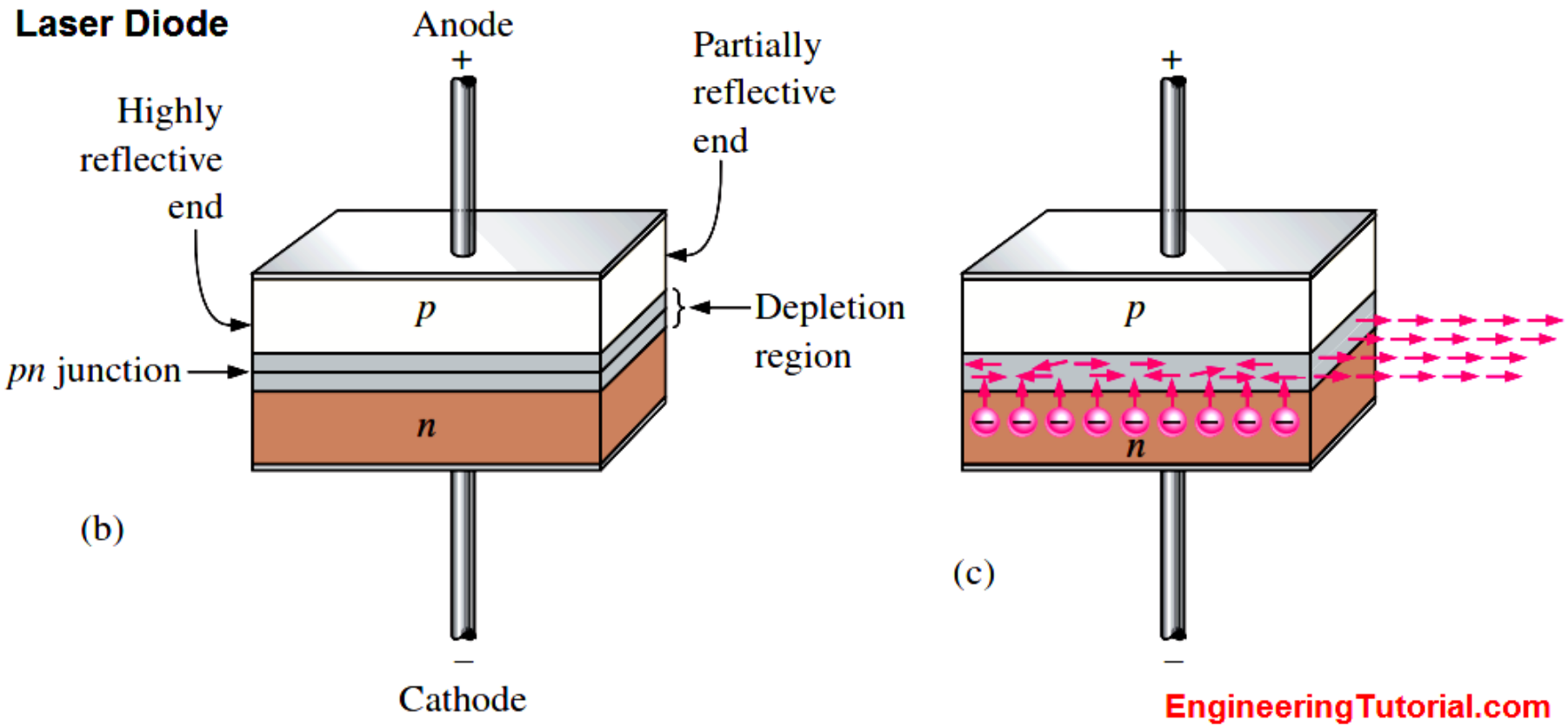
Types of Optoelectronics Device

Laser Diodes:

- When a voltage is applied across the P-N junction, the population inversion of the electrons is produced and then the laser beam is available from the semiconductor region
- The ends of the P-N junction of the laser diode have polished surface and hence, the emitted photons reflect back to create more electron pairs
- Thus, the photons generated will be in phase with the previous photons

Types of Optoelectronics Device

Laser Diodes:



Application of Optoelectronics Device

1. LEDs could become the next generation of lighting and used anywhere like in indication lights, computer components, medical devices, watches, instrument panels, switches, fiber-optic communication, consumer electronics, household appliances, traffic signals, automobile brake lights, 7 segment displays and inactive displays, and also used in different electronic and electrical engineering projects such as

   - Propeller Display of Message by Virtual LEDs
   - LED Based Automatic Emergency Light
   - Mains Operated LED Light
   - Display of Dialed Telephone Numbers on Seven Segment Display
   - Solar Powered Led Street Light with Auto Intensity Control

Application of Optoelectronics Device

2. The solar cells are applicable in rural electrification, telecommunication systems, ocean navigation aids, and electric power generation in space and remote monitoring and control systems and also used in different solar energy based projects such as

   - Solar Energy Measurement System
   - Arduino based Solar Street Light
   - Solar Powered Auto Irrigation System
   - Solar Power Charge Controller
   - Sun Tracking Solar Panel

Application of Optoelectronics Device

3. Photodiodes are used in many types of circuits and different applications such as cameras, medical instruments, safety equipment, industries, communication devices and industrial equipment

4. Optical fibers are used in telecommunications, sensors, fiber lasers, bio-medicals and in many other industries

5. The laser diodes are used in fiber optic communication, optical memories, military applications, CD players, surgical procedures, Local Area Networks, long distance communications, optical memories, fiber optic communications and in electrical projects such as RF Controlled Robotic Vehicle with Laser Beam Arrangement and so on


Thank
you